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Third-Generation Power Control Unit for Small Hybrid Vehicle※

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1. Introduction

Recently, stringent rules and regulations related to fuel consumption and exhaust gas have been imposed on automobiles worldwide to address the issues of global warming and atmospheric pollution. Many automobile manufacturers have urgently demonstrated a policy to increase the number of electric-drive vehicles. The Paris Agreement, which entered into force in 2016, sets a goal to limit the global temperature increase to within 2ºC above "pre-industrial level." To achieve this, according to an estimation announced by the International Energy Agency, more than 90% of automotive vehicles must be electric-drive by 2050[(1)]. For the purpose of the widespread usage of electric-drive vehicles, it is essential for the vehicles to provide performance and convenience equivalent to those of conventional gasoline-fueled vehicles and to be sold at an acceptable price to users. However, the use of electric-drive systems necessitates additional components such as batteries, e-motors, and power control units (PCUs), which leads to a reduction in the cabin space and an increase in cost. These issues are decelerating electrification and are more apparent in the case of small vehicles. Keihin has released a mass-production model of the second-generation PCU (GEN2-PCU) for mid-sized two-motor hybrid vehicles[(2),(3)]. The PCU is equipped with an all-in-one power module that was developed in-house. This article introduces the third-generation PCU (GEN3-PCU), which has a small size and low cost and is based on the GEN2-PCU. The GEN3-PCU was developed for small hybrid vehicles.

2. Product Overview

The newly developed GEN3-PCU includes two e-motors for traction and generation along with boosting functionality for adjusting the motor drive voltage and is intended to be installed in small-sized strong-hybrid vehicles that achieve low fuel consumption and high driving performance at the same time. The GEN3-PCU, therefore, has an inverter and boosting circuit to control the two e-motors, as well as a DC-DC converter for auxiliary devices on board. In addition to incorporating multiple converting functions, the GEN3-PCU is required to have a reduced size and cost in order to be installed in small-sized vehicles.

Figure 1 presents an external view of the GEN3-PCU, and Table 1, Fig. 2, and Fig. 3 show its

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Table 1 Comparison of specifications between GEN2 and GEN3

		GEN2	GEN3
Hybrid system	System maximum voltage [V]	700	570
	Motor maximum output torque [Nm]	315	253
	Motor maximum output power [kW]	135	80
PCU	Maximum total output power [kVA]	388	267
	Maximum boost voltage [V]	700	570
	Motor maximum output current [Arms]	320	270
	Generator maximum output current [Arms]	155	155
	DC-DC converter maximum output current [A]	-	140
	Volume [L]	8.9	8.5
	Weight [kg]	14.3	13.4

main specifications, a circuit block diagram, and an exploded view of components, respectively. The main circuit of the GEN3-PCU is composed of inverters for the traction motor and generator as well as the voltage control unit (VCU; composed of a reactor, a condenser, and power devices). The GEN3-PCU also features a gate-drive circuit and a control circuit

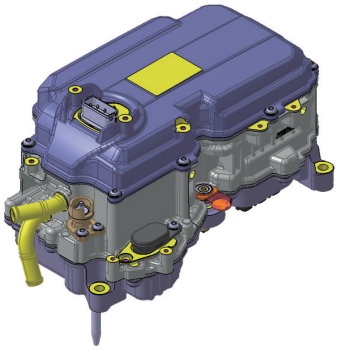


Fig. 1 External view of the GEN3-PCU

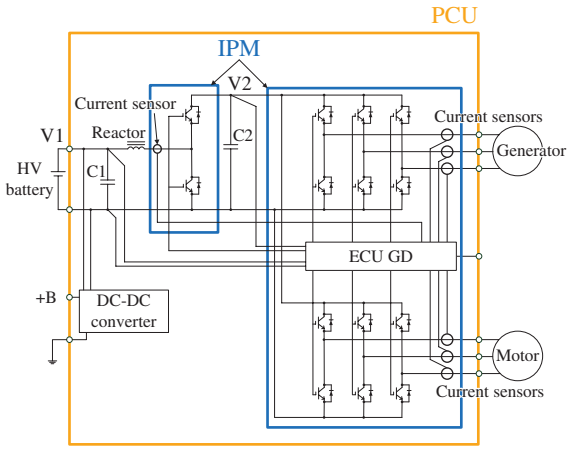


Fig. 2 Circuit block diagram of the GEN3-PCU

for controlling the main circuit, current sensors for detecting the three-phase output and current at the boost circuit, voltage sensors for high-voltage areas, and the DC-DC converter for auxiliary devices.

The GEN2-PCU and GEN3-PCU have similar main circuit and control circuit configurations. The motor output for the GEN3-PCU is smaller than that for the GEN2-PCU owing to the difference in the vehicle size, while the GEN3-PCU incorporates the DC-DC converter for auxiliary devices that is located near the battery in the case of hybrid vehicles in which the GEN2-PCU is installed. In addition, the GEN3-PCU is directly mounted onto the transmission case in the same manner as the GEN2-PCU.

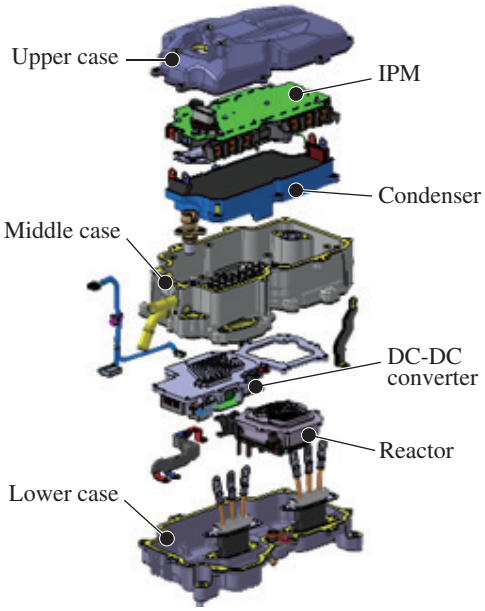


Fig. 3 Exploded view of the GEN3-PCU

3. Main Aims and Technological Aspects of Development

The GEN2-PCU is installed in a Category-C (medium class) vehicle, while the GEN3-PCU is to be used for a Category-B vehicle (small class). Thus, the development project aimed to make the GEN3-PCU small at a low cost.

The GEN3-PCU also adopts the all-in-one intelligent power module (IPM), which was

Table 2 Breakdown for parts of GEN2-PCU and GEN3-PCU

			GEN2	GEN3
IPM components	Power module		Integrated	Integrated
	Gate driver circuit		Integrated	Integrated
	Electric control unit		Separated	Integrated
	Current sensor		Separated	Integrated
	Discharge circuit		Separated	Integrated
Condenser	Capacitance [μF]	High voltage	1124	566
		Low voltage	236	340
Reactor	Inductance [μH at 0A]		273	184
	Resistance [mΩ at 20ºC]		19.3	9.7
DC-DC converter			Not incorporated	Incorporated
Number of parts on ECU/gate driver			2,200	1,300
Number of parts on power module			250	160
Number of PCU cases			4	3
Number of harnesses			17	3
Number of bus bars			4	2

originally developed for the GEN2-PCU by Keihin and is suited for small-sized PCUs.

The main characteristics of the GEN2 all-in-one power module are as follows: 1) The power devices for all main circuits for the motor drive inverter, generator inverter, and VCU are integrated into one power module; 2) The power devices are directly cooled without using materials with higher thermal resistance such as thermal compounds; 3) Instead of using the commonly used aluminium wire, lead frames made of thin copper plates are employed for the main circuit wiring connecting the emitters for the power devices and the external bus bars. These features helped to realize a compact power module with higher cooling capacity and thus a small-sized PCU.

In addition to employing the all-in-one IPM, the GEN3-PCU has been developed with the aim of downsizing at a low cost by introducing the following new technological aspects:

1. function-integrating all-in-one IPM;
2. smaller power devices with smaller losses;
3. smaller VCU with higher efficiency;
4. packaging for the smaller PCU; and
5. more efficient production processes

Table 2 compares the PCU components of GEN2 and GEN3. For the GEN3-PCU, components have been integrated and downsized and it has a significantly reduced number of components compared with the GEN2-PCU, which reduces the PCU size and cost. The following sections comprehensively discuss techniques introduced for the GEN3-PCU.

4. Function-Integrating All-in-One IPM

The function-integrating all-in-one IPM for GEN3 has its functionalities enhanced on the basis of the GEN2 all-in-one IPM. As shown in Fig. 4, the GEN2-IPM incorporates a high-power main circuit composed of a power semiconductor, a gate-drive function, and a voltage-sensing function. On the other hand, the GEN3-IPM includes a control function for gate-drive conditions that uses commands from the upper-level ECU, as well as a current-sensing function for monitoring the output current. From the hardware perspective, GEN3 integrates nine parts of GEN2, the power module and gate-drive circuit board (gate driver: GD), the current sensor assembly which is composed of three modules and harnesses, the control circuit board (electrical control unit: ECU), a constant discharge resistor, a panel lock connector and harness assembly and harnesses between the ECU and the GD, as shown in Fig. 5. In addition, the adoption of new IC and integration/reduction of circuit functionalities reduces the number of parts to be mounted onto the gate-driver board and the ECU by approximately 40%. Moreover, the function integration results in an electrical connection among functionalities without using harnesses and

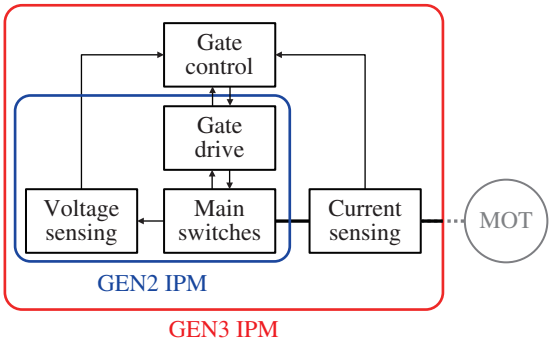


Fig. 4 Comparison of integrated functions in IPM between GEN2 and GEN3

thus increases the noise resistance. This reduces the number and the constant value of noise filters, increasing the circuit response performance. More specifically, the accuracy and response performance related to sensing for current, voltage, and temperature have been increased to improve controllability.

The major technological points for realizing the GEN3-IPM are as follows:

1. power modules suited for function integration and circuit board design;
2. downsizing technology for circuit boards; and
3. integration technology for the current sensor

4.1. Power modules suited for function integration and circuit board design

Figure 6 shows the configurations of the power modules and the circuit boards. In the GEN3 power module configuration, the high-voltage DC bus line is located along the center line between the upper and lower arms of the main inverter circuit, with the magnetic cores for the current sensors on the outermost areas. Conversely, the circuit board configuration has the central low-voltage control circuit sandwiched by high-voltage drive circuits that are surrounded by current-sensing circuits including Hall ICs for the current sensors.

In addition, the low-voltage external interface parts are situated on one side of the circuit board, while high-voltage-sensing circuit and discharge circuit are situated on the other side. This configuration makes it possible to realize a compact IPM without compromising the multiple functionalities on the circuit board.

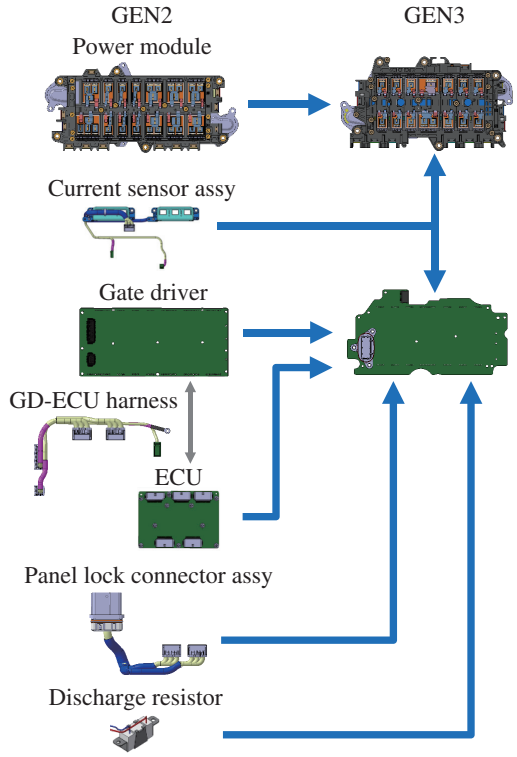


Fig. 5 Integrated components in the GEN3-IPM

4.2. Downsizing technology for circuit boards

The circuit configuration explained in 4.1 requires a reduction in the number of parts and an increase in the mounting density to integrate the control circuit, gate-drive circuit, current sensor circuit, and other related devices in the limited area of the circuit board. For example, the voltage-sensing circuit for GEN2 consists of a resistor array and operational amplifiers. The number of parts and mounting area of 180 and 2,500 mm^2 in GEN2 have been significantly reduced to approximately

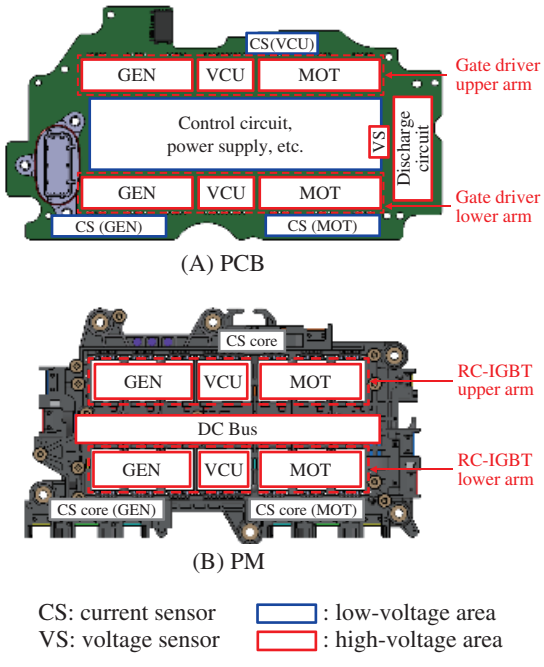


Fig. 6 Configurations of PM and PCB in the GEN3-IPM

18 and 270 mm^2, respectively, in the case of GEN3 as a result of the use of IC configuration for these parts (Fig. 7). Furthermore, multiple power supply units for the gate drive of each arm, as well as those for the control circuit and gate drive, have been integrated. The integration of the control circuit, gate-drive circuit, and current sensor circuit significantly reduces the number of noise filter circuits and connectors (Fig. 8). From the viewpoint of mounting technology, the high reliability demanded for on-board parts is balanced with the reduced mounting area, due to the reduced insulation distances and dense mount soldering techniques.

4.3. Integration technology for the current sensor

As shown in Fig. 2, the GEN3-PCU has current sensors corresponding to seven phases. The GEN2-PCU current sensor is composed of three current sensor assemblies for three phases of the e-motor, three phases of the generator, and one phase for the VCU along with connection harnesses between ECUs. These parts are integrated into the IPM in GEN3 (Fig. 9). Figure 10 presents the cross-sectional view of the current sensor integrated into the IPM. Magnetic cores constituting the current sensor are arranged around the bus bar in the power module case, whereas the Hall ICs are mounted on the PCB. This configuration integrates current sensors into

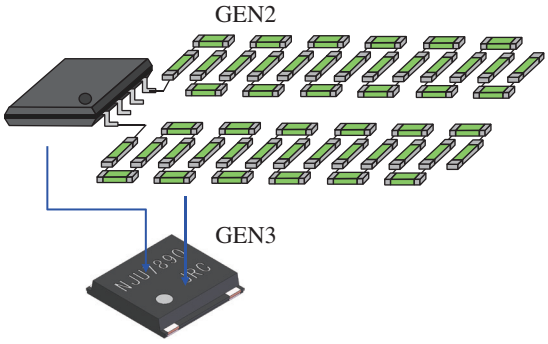


Fig. 7 Integration of voltage sensing circuit

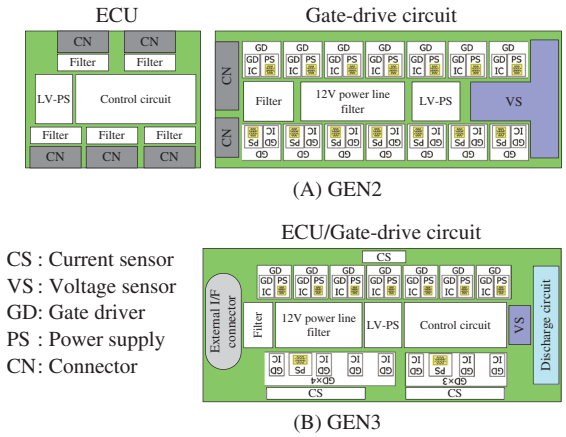


Fig. 8 Comparison of component layouts on PCB between GEN2 and GEN3

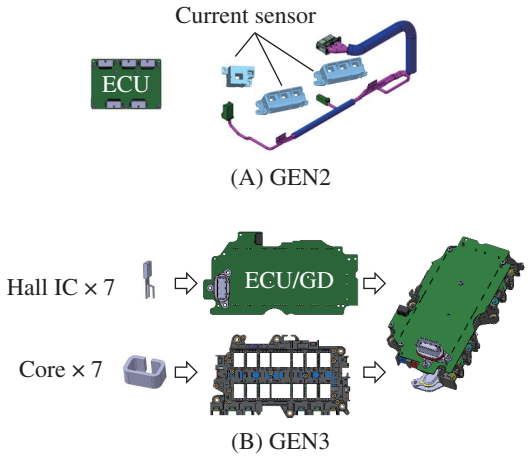


Fig. 9 Comparison of current sensor configuration between GEN2 and GEN3

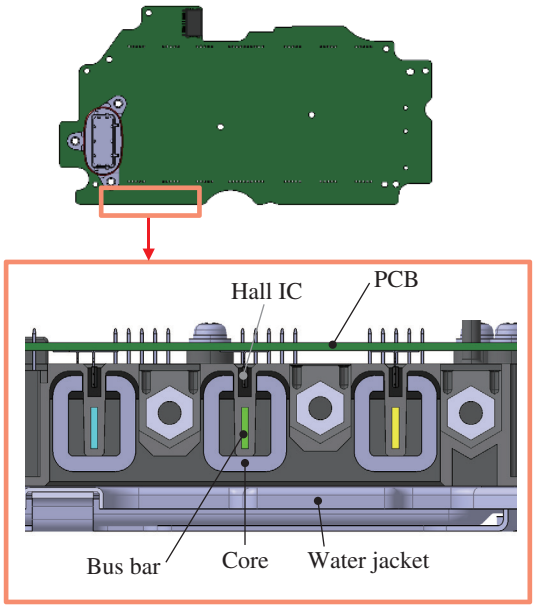


Fig. 10 Cross-sectional view of integrated current sensor

the IPM and eliminates the harnesses, the assembly cases and the PCBs for the sensors used in the GEN2, making it possible to significantly reduce the size and number of parts.

The challenges when realizing this current sensor integrated into the IPM are as follows: 1) heat exposure in the operation environment and during the manufacturing processes and 2) characteristic correction in the integrated condition into the IPM.

The current sensors are exposed to a wide range of operating temperatures, and the characteristics of the current sensors are considerably affected by the change in the core gap length due to stress originating in the temperature change. In addition, as shown in the manufacturing process flow in Fig. 11, magnetic cores are exposed to high temperatures in the reflow process for soldering other parts after they are mounted on the power module case. Care should be taken to prevent structural failures such as the detachment of core fixation because of the effect of the heat stress. Therefore, the material and structure for fixing magnet cores onto the power module case are designed to suppress the change in core gap length due to the temperature change in the operating environment and to increase the resistance against the heat stress during the reflow process.

Moreover, the Hall ICs are mounted onto the PCB in the part surface mounting process. Following this, the structures of the current sensors are completed in the IPM process where the power module and the PCB are assembled. To increase the accuracy of the current sensors, the variations of individual magnetic cores and Hall ICs for detecting magnetic fields must be corrected. This correction can be carried out via gain correction that applies current after the current sensor is completed. As such, the GEN3-PCU includes a circuit for gain correction on the PCB, enabling gain correction of the IPM components.

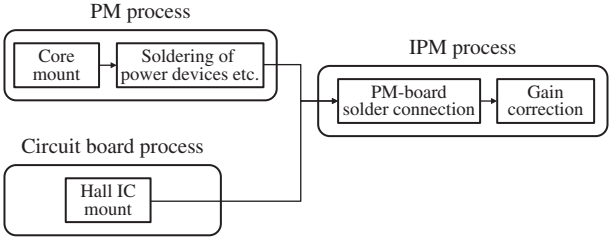


Fig. 11 Production flow of integrated current sensor

5. Small-Sized Power Device with Low Losses

Conventional switching circuits for on-board motor drive inverters employ insulated gate bipolar transistors (IGBTs) and rectifying diodes, which are two-terminal elements. Power semiconductor devices are essential and costly components that significantly affect the output power and PCU efficiency. Costs can be reduced by reducing the device area; however, this increases losses and thus reduces the maximum output power. The question is how to reduce the device area with a good balance with the required output power and efficiency. To achieve this, the following approaches were taken in the project:

1. the use of RC-IGBTs;
2. a reduction in power device losses;
3. high-temperature operation; and
4. more accurate and rapid protection functions for devices

As a result, apart from the difference in vehicle class between GEN2 and GEN3, the area of power devices of GEN3 has been reduced to less than half when compared to that of GEN2 (Fig. 12).

5.1. Use of RC-IGBTs

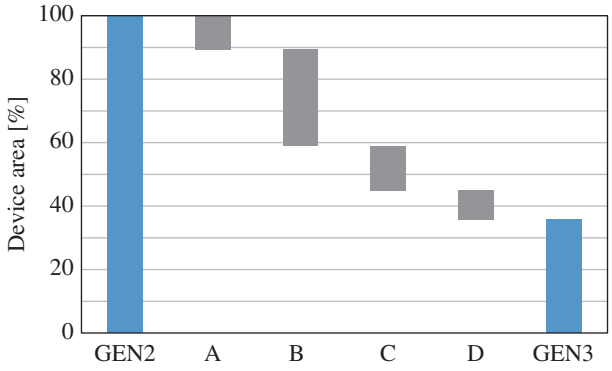
A reverse-conducting IGBT (RC-IGBT) integrates an IGBT and a diode. Figure 13 shows the top view and schematic cross-sectional view of an RC-IGBT, while Fig. 14 shows the top view of layout structures with the power devices and related parts when mounted

for one arm to be installed on the power module. If the widely used combination of an IGBT and a diode is replaced with an RC-IGBT, great benefits can be expected in terms of a reduced device area, small-sized power module, and reduced number of mounted parts.

We shall now discuss the reduced device area. In the configuration of the conventional combination of an IGBT and a diode, the temperature-sensing diode for devices is provided solely for the IGBT. Therefore, the diode area should include margins because the temperature of a diode cannot be directly measured. On the other hand, the use of an RC-IGBT can considerably reduce the diode area because it can detect the temperatures of both the IGBT and the diode using a sensing diode. As can be seen from Fig. 15, switching losses of the diode can be reduced by reducing the device area. Thus, the use of RC-IGBTs can reduce switching losses.

In addition, in the VCU, there are many modes where either the IGBT or the diode is activated. The heat resistance can be reduced because the heat generated by power devices is dissipated from the surfaces of both the IGBT and the diode, contributing to a reduced area for devices (Fig. 16).

In addition, owing to the one-chip configuration, the use of RC-IGBTs can reduce the area for the guard rings that, in the case of the conventional combination of an IGBT and a diode, are used to



A: Specification change from GEN2 to GEN3
B: Power loss reduction in RC-IGBT
C: High-temperature operation
D: Use of RC-IGBT

Fig. 12 Comparison of power device areas between GEN2 and GEN3

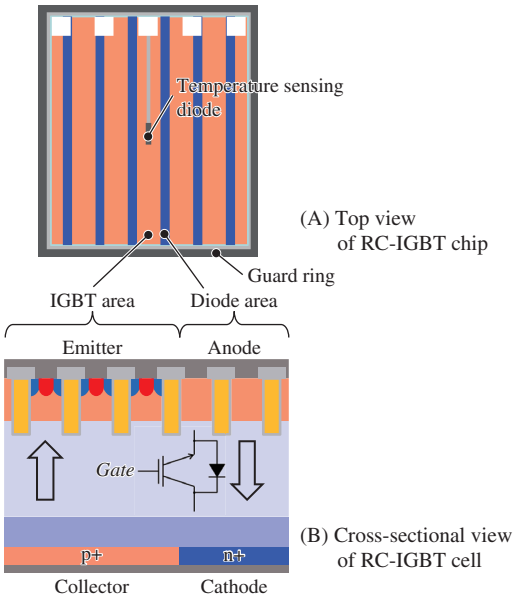
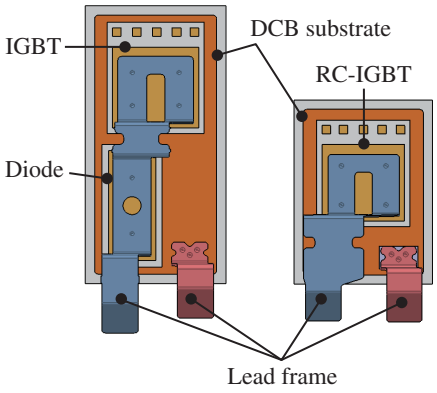


Fig. 13 Top view and schematic cross-sectional view of RC-IGBT



(A) IGBT/Diode pair (B) RC-IGBT

Fig. 14 Layout structures of power devices and related parts in PM

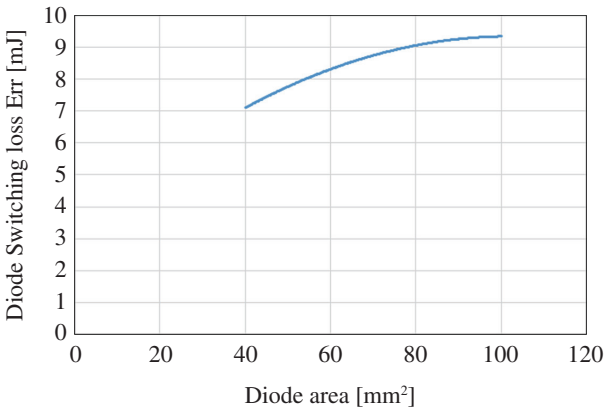


Fig. 15 Diode switching loss vs. device area

maintain the voltage resistance performance around the outer circumference of each device. These measures reduce the area by approximately 10% compared with that of the combination of an IGBT and a diode.

In addition to the significant reduction in the number of devices, the use of an RC-IGBT also allows a reduction in the amount of solder and the size of the lead frame and DCB board (Table 2, Fig. 14). As a result, the power module could be downsized at lower costs.

5.2. Reduction in power device losses

Losses of power devices can be reduced by the following: 1) enhancing power device characteristics; and 2) reducing the power loads (current, voltage, and switching frequency) applied to power devices. GEN3 employs the latest generation device to reduce the losses. Efforts were made to reduce the electric loads applied to power devices as much as possible within the range of the required vehicle driving performance. Figure 17 shows a schematic diagram of the VCU. It controls voltage and current to generate motor torque at corresponding rotation speed according to the demands from the upper-level ECU. Optimization of operation as a hybrid vehicle including an internal combustion engine involves optimizing the voltage and current applied to the e-motor, as well as the switching frequency, resulting in output characteristics balanced with reduced losses. As a result, the device area has been reduced by approximately 30%.

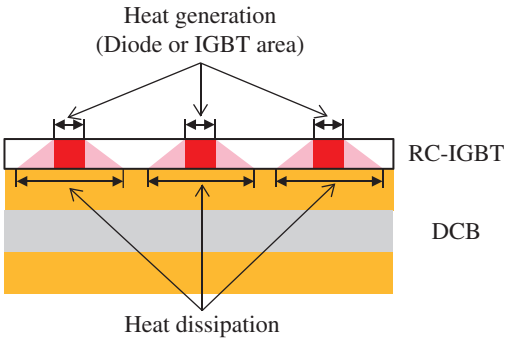


Fig. 16 Heat generation and dissipation in RC-IGBT

5.3. High-temperature operation

High-temperature operation of power devices is also effective in reducing the area used for power devices. Feasible measures for high-temperature operation include an increase in the allowable maximum operating temperature of the device and more accurate and rapid detection of the power device temperature. The temperature is detected using a temperature-sensing diode situated at the center of the RC-IGBT. The GEN3-PCU successfully increases the detection accuracy and speed by integrating the ECU with the gate-drive circuit and reading temperature sensor data from the diode in a digital format, accomplishing a temperature increase

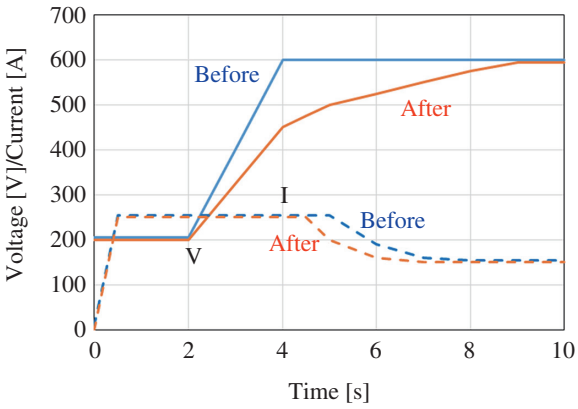


Fig. 17 Suppression of heat generation during the VCU operation

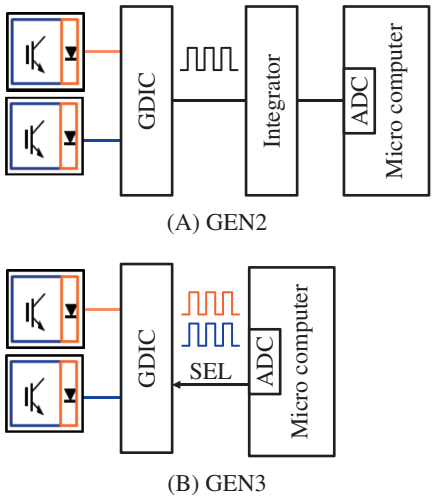


Fig. 18 Comparison of the device temperature sensing circuit between GEN2 and GEN3

in device operation of 5ºC or more. Furthermore, two RC-IGBTs are used in parallel in the lower arm of the VCU. The characteristics of both temperature-sensing diodes are stored in a microcomputer, and the temperatures of both devices are continuously measured. By doing this, the detection accuracy is increased by 5ºC or more. This high-temperature operation approach has been used to reduce the device area by 10% or more.

5.4. More accurate and rapid protection function for devices

As discussed above, the GEN3-PCU has a reduced device area because of a reduction in the device losses and an increase in the operating temperature. However, the device area may be restricted due to the safe operation area instead of the device temperature. To respond to this case, one effective measure is to accelerate the protection operation by increasing the detection speed for current and voltage. The GEN3-PCU integrates the current sensor, gate-drive circuit, and control circuit to enhance the circuit response performance. In the case of protection operations for overcurrent, the delay time (T_{delay}) until the RC-IGBT is shut down after detecting an overcurrent is reduced to less than half. As a result, the overcurrent value is reduced by 200 A for example (Fig. 19), contributing to an unrestricted state of the device area because of the safe operation area.

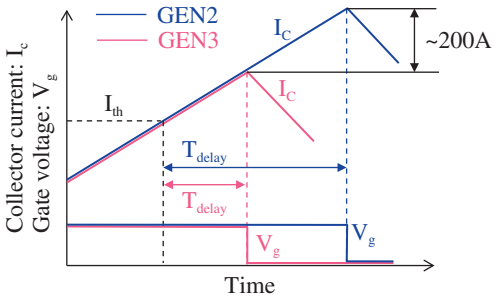


Fig. 19 Schematic waveforms in overcurrent protection operations of GEN2 and GEN3

6. Reducing Size and Increasing Efficiency of VCU

The circuit diagram of the VCU is shown in Fig. 20. During the development of the GEN3-VCU, the main focus was suppressing the power losses of the reactor and the RC-IGBT and downsizing the condenser and the reactor, which occupied a large volume, without failing to provide the desired voltage and current control characteristics.

Most of the power losses of the VCU consist of conduction losses of the reactor wiring, iron losses of the magnetic cores, and conduction losses and switching losses of the power semiconductor switch. In addition, the VCU is required to control the ripple current and voltage below a certain level. These losses and ripple current vary depending on the resistance of the reactor wiring and VCU operating frequency (Fig. 21). The

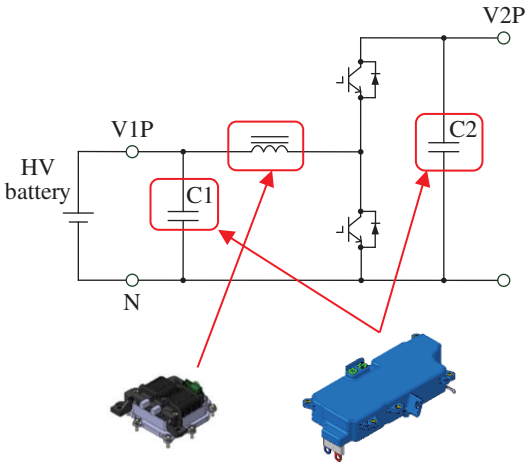


Fig. 20 Circuit diagram and components of the VCU

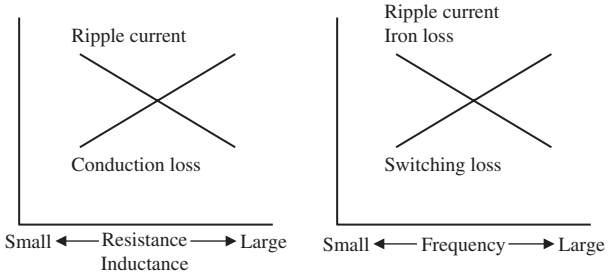


Fig. 21 Ripple and loss dependence on the VCU parameters

magnitude relationship of these values also changes according to the VCU operation points. The GEN3-VCU reduces the copper losses by reducing the wiring resistance value of the reactor to less than a half compared with that of the GEN2-VCU and determines the proper switching frequency for each VCU operation mode. These measures reduced the losses of the entire VCU. The reactor size has thus been reduced by approximately 20%.

Moreover, the capacitance of the condenser must be reduced for downsizing the condenser, whereas a reduction in the capacitance may increase the voltage fluctuation degree due to the resonance of the impedance of the PCU and external loads. An associated issue is the excessive temperature rise resulting from the increased current density for the condenser. These two issues must be solved to reduce the capacitance and the condenser size. Newly developed control functions to address the issue of an increase in the voltage fluctuation degree are current control and voltage control by way of cascade control, disturbance observer, duty estimation operator, and duty compensation operator (Fig. 22). The application of these control methods has suppressed the voltage resonance gain in a broad range of frequency and reduced the AC components of the load current (Fig. 23). To suppress the temperature rise, the condensers were located between the upper and lower cooling water channels, as to be discussed in the next section (Fig. 24). These measures reduced the condenser volume almost by half as compared with the GEN2-PCU.

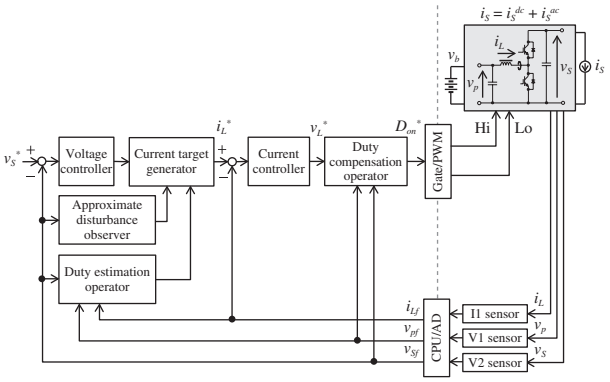


Fig. 22 Control block diagram of ripple suppression during the VCU operation

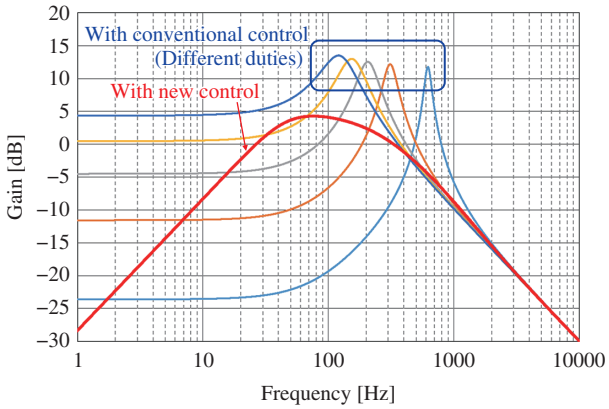


Fig. 23 Magnitude plots with and without applying ripple suppression

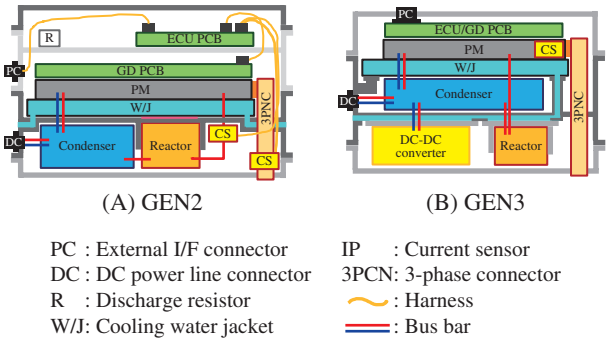


Fig. 24 Comparison of schematic configurations of PCU between GEN2 and GEN3

7. Downsizing Packaging

The GEN3-PCU is directly mounted onto the transmission case in the engine compartment as in the case of the GEN2-PCU. More severe packaging restrictions have been imposed on the GEN3-PCU because it is used for small-sized hybrid vehicles and incorporates a DC-DC converter, unlike the GEN2-PCU (Fig. 25). The DC-DC converter supplies power to 12 V electric devices on board and is designed for an output of 2.1 kW at the maximum at a high efficiency under severe conditions in terms

of space, heat, and vibration.

Figure 24 presents schematic cross-sectional configurations for the GEN2-PCU and the GEN3-PCU. When directly mounting the PCU onto the transmission case, heat and vibration transmitted from the engine and e-motor to the PCU should be considered. In terms of heat, the main heat-generation components such as the power modules, reactor, and DC-DC converter are arranged to be directly cooled, while the condensers are situated between two cooling water channels to keep the surrounding temperature low. This sophisticated cooling structure successfully facilitates the reduction in size of the condensers, the reactor and the DC-DC converter. With respect to vibration, the design incorporates, firstly, a low center of gravity for the PCU, along with separation of the vibration frequency transmitted from the engine and the e-motor to the PCU, and the eigenfrequencies of the individual parts of the PCU, a stress mitigation structure, and structures giving reinforced part fixture.

Along with the above points, the distance between the power modules and high-voltage condensers has been shortened to suppress the surge voltage.

The GEN3-PCU has reduced size and weight even though it incorporates the DC-DC converter, owing to the above-discussed items of heat, vibration, and electricity, along with a significantly reduced number and size of parts, owing to the abovementioned functionality integration with the IPM, the use of RC-IGBT, and new control methods for the VCU (Table 2).

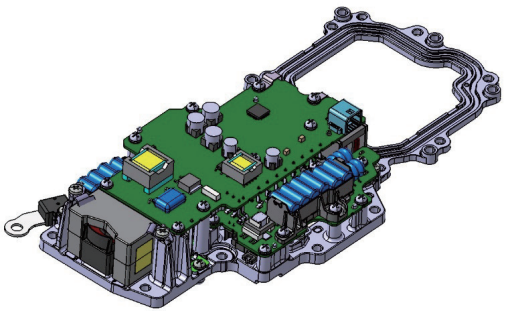


Fig. 25 External view of the DC-DC converter

8. Highly-Efficient Production Processes

The GEN3-PCU has a considerably reduced number of parts, including harnesses that have to be assembled manually, reducing the processes for production and contributing to automation.

The GEN3-PCU reduces the reflow processes for soldering internal parts of power modules to once, as compared to twice in the case of the GEN2-PCU (Fig. 26). Soldering is applied to more than 70 locations in the power devices, the lead frame, the insulation boards, the cooling water jacket, the bus bars and other components in the power module case (this is compared to 110 locations in the GEN2-PCU). In consideration of the yield rate, in the case of the GEN2-PCU, soldering for the sub-assembly composed of the power devices, the insulation boards, and the lead frame was applied in the first reflow, and following the sub-assembly inspection procedure, soldering for the other parts was performed in the second reflow. The technique learned in the mass-production experience of the GEN2-PCU is utilized in the GEN3-PCU to modify the power device inspection process before reflow, and the reflow process significantly reduces the area of power devices. As a result, the reflow processes are unified.

These approaches significantly increase the efficiency of GEN3-PCU production processes and thus reduce costs.

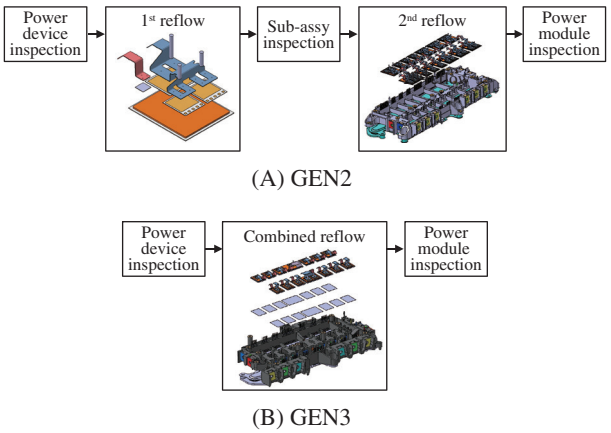


Fig. 26 Comparison of power module production processes between GEN2 and GEN3

9. Conclusion

The GEN3-PCU has newly been developed for a small-sized two-motor hybrid vehicle. The PCU is required to have a reduced size and cost because it is used for small vehicles; therefore, the PCU is based on the all-in-one IPM for the GEN2-PCU; its components are integrated; the size and number of components are reduced; and the production processes efficiency is increased, by means such as reducing the processes for production and promoting automation. As a result, a small-sized and lightweight PCU including a DC-DC converter and with low cost compared to the GEN2-PCU has been achieved.

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We have developed a product that contributes to the realization of the upcoming society embracing full-fledged vehicle electrification. We express our sincere thanks to Honda R&D Center and Honda Motor Co., Ltd., our parts suppliers, and our colleagues involved, who provided assistance in various fields, starting from the development of elemental technology until mass production. (NONAKA)